

## Assignment 3

### Parallel and Distributive Computing (CS 3006)

#### Instructions:

- Each file that you submit must contain your name, student-id e.g. make ROLL-NUM\_SECTION\_A2.cpp (23i-0001\_A\_A2.c) and so on., and assignment # on top of the file in comments.
- All the submissions will be on Google classroom within the deadline. Submissions other than Google classroom (e.g. email etc.) will not be accepted.
- The student is solely responsible to check the final zip files for issues like corrupt files, viruses in the file, mistakenly exe sent. If we cannot download the file from Google classroom due to any reason, it will lead to zero marks in the assignment.
- Displayed output should be well mannered and well presented. Use appropriate comments and indentation in your source code.
- Be prepared for viva or anything else after the submission of the assignment.
- If there is a syntax error in code, zero marks will be awarded in that part of assignment.
- Understanding the assignment is also part of the assignment.
- Zero marks will be awarded to the students involved in plagiarism. (Copying from the internet is the easiest way to get caught).
- Late Submission policy will be applied as described in the course outline.

**Tip:** For timely completion of the assignment, start as early as possible.

**Note:** Follow the given instruction to the letter, failing to do so will result in a zero.

**[CLO 2 : Implement different parallel and distributed programming paradigms and algorithms using Message-Passing Interface (MPI) and OpenMP ]**

### **Scalable PageRank Engine & Communication Architecture Benchmarking:**

In this assignment, you will design, implement, and rigorously analyze a distributed-memory version of the PageRank algorithm using MPI. You will leverage **ParMETIS** to perform parallel graph partitioning, ensuring that your data distribution minimizes cross-process communication. You will then evaluate the performance trade-offs between Point-to-Point, Collective, and Asynchronous Overlap communication scenarios.

#### **The PageRank Power Method:**

The objective is to discover the stationary distribution of a random walk across a directed graph. In each iteration  $k$ , the rank of vertex  $i$  is calculated using the following probability model:

$$PR_{k+1}(i) = \frac{1-d}{N} + d \sum_{j \in In(i)} \frac{PR_k(j)}{L(j)}$$

#### **Parameters & Mathematical Constraints:**

- **Damping Factor ( $d$ ):** Fixed at **0.85**. This represents the probability that a "surfer" continues clicking links rather than jumping to a random page.
- **Convergence:** Your engine must iterate until the global  $L_1$  norm of the residual (the difference between  $PR_{k+1}$  and  $PR_k$  falls below a threshold  $\tau = 10^{-7}$

This video link will help you to understand Page Rank algorithm:

<https://www.youtube.com/watch?v=C4ESsZWWmKU>

#### **The METIS Partitioning Flow**

**ParMETIS** is an MPI-based parallel library designed for high-performance graph partitioning and repartitioning. Rather than using a simple linear distribution, you must utilize **ParMETIS** to partition the graph data across your MPI ranks. Your implementation must follow this roadmap:

1. **Construct** the initial graph structure required by ParMETIS (typically an adjacency list/CSR-like format)

**Note:** You can use any external tools /custom scripts or libraries to convert provided dataset into acceptable input format for METIS or use any other pre converted data available. (this won't be part of evaluation)

## 2. Graph Partitioning with ParMETIS:

- Compute a high-quality partition that minimizes the "edge-cut" (the number of edges crossing between different MPI ranks).
- Redistribute your local graph data based on the partition IDs returned by ParMETIS.

## 3. Local Graph Representation: Once the data is redistributed, each process stores its local graph data.

### Vertex Categorization for Optimized Communication:

To implement communication scenarios, you must classify your local vertex set based on the partitioning generated by ParMETIS:

1. **Internal Vertices:** Nodes whose incoming links originate entirely from vertices owned by the same MPI rank.
2. **Boundary Vertices:** Nodes that require data from a remote MPI rank to complete their PageRank calculation.
3. **Ghost Cells :** Local memory buffers that store the "stale" PageRank values of remote vertices required for Boundary Vertex calculations.

### Implementation Scenarios: The Comparative Study

#### Scenario 1: Manual P2P Distribution

- Mechanism: Use MPI\_Send and MPI\_Recv.
- Rank 0 distributes graph slices using a manual handshake protocol (send size first, then data). Iterations use blocking

#### Scenario 2: Collective Operations

- Mechanism: Use MPI\_Scatterv and MPI\_Gatherv.
- Use MPI\_Scatterv and MPI\_Allgatherv (justify why Allgatherv is preferred for synchronization). Focus on managing displacements (displs) and global barrier overhead.

**Note:** Since the global rank vector is needed by all processes to compute the next iteration, consider whether MPI\_Allgatherv is more appropriate than Gatherv followed by a broadcast.

### **Scenario 3: Communication-Computation Overlap**

- Mechanism: Use MPI\_Isend, MPI\_Irecv, and MPI\_Waitall.
- Use non-blocking MPI\_Isend, MPI\_Irecv, and MPI\_Waitall to hide communication latency behind the computation of Internal Vertices.

### **Analysis Report Requirements:**

Your final report must feature:

- **Strong Scaling Plots:** Compare execution time for Scenario 1,2,3 across 2, 4, 8, and 16 cores.
- **Scenario Comparison:** A bar chart comparing the "Time per Iteration" for Scenarios 1, 2, and 3.
- **Overlap Efficiency:** Quantitative comparison of your "Overlap" version vs. a "Sequential Communication" version, calculating the percentage of communication time successfully "hidden."

### **General Guideline Steps:**

1. You may utilize any external tools, custom scripts, or libraries to convert the provided edge-list datasets into the METIS-compatible format. Alternatively, you may use the pre-converted datasets available.
2. Load the graph data and invoke ParMETIS\_V3\_PartKway to assign vertex ownership across ranks while minimizing the "edge-cut." You must then implement a redistribution routine to move graph slices to their assigned MPI ranks based on these partition IDs.
3. Classify local nodes into Internal and Boundary vertices, using Ghost Cells to buffer remote PageRank values. Implement the Power Method with a damping factor of 0.85 and iterate until the global  $L_1$  norm of the residual falls below  $10^{-7}$ .
4. Develop three versions of the engine: one using P2P (MPI\_Send/Recv), one using Collectives (MPI\_Allgatherv), and a high-performance version using Asynchronous Overlap (MPI\_Isend/Irecv).
5. Design performance analysis report.

**Note to Students:**

- **METIS Integration:** You are expected to link your project against the ParMETIS library.

<https://github.com/KarypisLab/ParMETIS>

- **Dataset:** Validate your engine on the [web-Google](#) dataset (875k nodes, 5.1M edges).

**Submission:**

- You have to submit report in word or pdf format.
- You have to provide the project source code in .zip format which you have used in your assignment.

### Rubrics :

| Category               | Key Requirements                                                                                                                                     | Points Distribution |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Functional Correctness | <b>15 pts:</b> Accurate PageRank power method and convergence.<br><b>15 pts:</b> Successful validation & correct ranking on dataset.                 | <b>30</b>           |
| Data & Partitioning    | <b>10 pts:</b> Correct METIS integration & edge-cut minimization.<br><b>10 pts:</b> Data redistribution & Internal/Boundary/Ghost cell logic.        | <b>20</b>           |
| MPI Scenarios          | <b>5 pts:</b> Scenario 1 (P2P Blocking).<br><b>10 pts:</b> Scenario 2 (Collectives/Allgatherv).<br><b>15 pts:</b> Scenario 3 (Asynchronous Overlap). | <b>30</b>           |
| Analysis Report        | <b>10 pts:</b> Scaling & Comparison plots.<br><b>10 pts:</b> Deep dive analysis on overlap efficiency & edge-cut impact.                             | <b>20</b>           |
| Total Score            |                                                                                                                                                      | <b>100</b>          |